

HADDIYA

BACKEND × ZID INTEGRATION

Backend Development Roadmap

Module-wise Integration Plan with ZID Platform

Document Scope	Backend Services + ZID Platform Integration
Frontend Status	Already designed and built (out of scope here)
Prepared for	Executive Leadership, Engineering, Stakeholders
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Table of Contents

1. Project Scope & ZID Platform Overview
2. Integration Architecture & Principles
3. Backend Development Timeframe
4. Team Composition (Backend-Only)
5. Module-wise ZID Integration Plan
6. Phase-wise Delivery Plan
7. Technical Architecture Stack
8. Risk Analysis
9. Final Recommendation

1. Project Scope & ZID Platform Overview

1.1 Backend Scope

Haddiya backend is the orchestration and intelligence layer sitting between the consumer apps (already designed and built in React + React Native) and the ZID e-commerce platform. ZID — the leading Saudi multi-tenant commerce platform — provides the core commerce primitives (catalog, orders, customers, payments, shipping). Haddiya backend extends ZID with multi-vendor marketplace logic, the Gift AI Assistant, send-as-gift workflows, scheduled delivery, loyalty, vendor portal APIs, and admin operations — all the differentiators that ZID alone does not provide out of the box.

1.2 Why ZID?

ZID is purpose-built for the GCC market with native support for Mada, STC Pay, Tabby, Tamara, Apple Pay, Arabic-first storefront APIs, local shipping carriers (SMSA, Aramex, Naqel), and PDPL-compliant data residency. Building Haddiya on ZID compresses 4–6 months of commerce primitives work and inherits proven payment and tax flows. Haddiya focuses engineering effort on the high-value differentiators — multi-vendor orchestration, AI, and the gifting experience.

1.3 Business Objectives

#	Objective	Target
1	Use ZID for commerce primitives, focus team on differentiators	Save 4–6 months of build time
2	Webhook-driven near-real-time sync across all entities	Sync lag < 5s p95
3	Idempotent integration that survives ZID API outages	Zero duplicate orders, zero lost webhooks
4	Vendor payout accuracy across multi-vendor splits	Reconciliation variance < 0.1% monthly
5	Backend SLA suitable for marketplace volumes	99.95% uptime, p95 API latency < 250 ms

1.4 Backend System Components

Component	Description
ZID Platform	Catalog, orders, customers, payments, shipping (system-of-record)
Haddiya BFF	REST + GraphQL gateway for mobile/web; routes to ZID and internal services
ZID Connector Service	Typed SDK wrapper, OAuth token mgmt, rate-limit handling, retries
Webhook Ingestion Service	Receives ZID events, validates HMAC, dispatches to internal queues
Catalog Cache Service	PostgreSQL + OpenSearch mirror of ZID products + categories
Multi-Vendor Orchestrator	Maps Haddiya carts to per-vendor ZID orders; payout split logic
Gift AI Service	LLM + pgvector embeddings; product recommendation API
Gift Experience Service	Message cards, scheduled delivery, recipient mgmt, gift wrapping
Loyalty Ledger	Points earn/redeem, tier rules, ZID coupon bridge
Vendor Portal API	Per-vendor scoped APIs built on top of ZID per-store endpoints
Admin Operations API	Approvals, disputes, refunds (proxied to ZID)
Notification Service	Push (FCM/APNs), WhatsApp Business, transactional email
Reporting Service	Data warehouse (Snowflake/BigQuery) fed from ZID + Haddiya ledger

Component	Description
Platform Foundation	PostgreSQL, Redis, OpenSearch, SQS, S3, CloudFront, WAF

2. Integration Architecture & Principles

Haddiya backend sits between the consumer apps and ZID. ZID is the system-of-record for all commerce primitives; Haddiya extends with marketplace, AI, and gifting logic. Six principles guide every integration decision:

Principle	Description
ZID owns commerce primitives	Products, orders, payments, shipping, customers live in ZID as system-of-record. Haddiya never duplicates them as writeable.
Haddiya caches for performance	ZID data mirrored to PostgreSQL + OpenSearch for fast queries, AI ranking, and reporting. Cache invalidated via ZID webhooks.
Haddiya owns differentiators	Multi-vendor mapping, Gift AI, scheduled delivery, message cards, loyalty ledger, vendor payout tracking — Haddiya tables only.
Webhooks-first sync	Real-time consistency via ZID webhooks; periodic reconciliation as safety net (every 15 min for inventory, hourly for orders).
API contract isolation	Mobile/web apps call Haddiya BFF, never ZID directly. Lets us swap ZID later without touching clients.
Idempotent everything	Every ZID write (order create, refund, customer update) is idempotent with deterministic external_id to survive retries.

2.1 High-level Data Flow

Consumer App → Haddiya BFF → Haddiya Services ↔ ZID Connector ↔ ZID Platform

- Reads: Mobile/web → BFF → Haddiya cache (Postgres + OpenSearch); cache hydrated from ZID via webhooks.
- Writes: Checkout, refunds, customer updates → Haddiya services → ZID API (with idempotency keys).
- Events: ZID webhooks → ingestion service → SQS → consumer workers (cache, ledger, notifications).
- Cron / safety net: Periodic reconciliation jobs catch any webhook misses.

3. Backend Development Timeframe

Frontend is already designed and built. These estimates cover backend services + ZID integration only. Buffer (~15%) included.

Release	Scope	Weeks	Months	Notes
Backend MVP	ZID account, OAuth, catalog sync, order orchestration, single-vendor checkout, single payment, basic webhooks	8–10	2–2.5	Pilot with 5–10 vendors
Backend V1 Production	Multi-vendor splits, full payment mix, send-as-gift backend, scheduled delivery, vendor portal API, admin API, Gift AI v1, full notifications	16–20	4–5	Public launch, ZID partner app live
Backend V2 Enterprise	Multi-country (UAE/Bahrain ZID stores), loyalty ledger, B2B gifting API, advanced analytics, anomaly detection, multi-warehouse	28–34	7–8.5	GCC expansion, enterprise SLAs
End-to-end (overlapped)	Phases run in parallel after MVP foundation	~26	~6.5	Backend complete with frontend already in place

Total cumulative duration (sequential): ~24 weeks / 6 months.

With parallel pods: ~22 weeks / 5 months to V1 production.

4. Team Composition (Backend-Only)

4.1 Recommended V1 Backend Team

Role	HC	Engagement	Duration	Notes
Backend Lead / Architect	1	Full-time	Full project	Owns ZID integration design, code review
Backend Engineers	3	Full-time	Full project	ZID SDK, sync pipelines, vendor services
AI / ML Engineer	1	FT Phase D, PT after	~3 months	Gift AI Assistant, embeddings
Data Engineer	1	PT (50%) from Phase F	~2 months	Analytics pipeline, warehouse
DevOps / Platform Engineer	1	Full-time	Full project	IaC, CI/CD, observability, on-call
Security Engineer	1	Part-time (40%)	From Phase B	OWASP, secrets, pen test
QA / Integration Engineer	1	Full-time	From Phase B	API contract tests, ZID sandbox QA
Product Manager	1	Full-time	Full project	Roadmap, vendor liaison, ZID partner mgmt
Tech Project Manager	1	Full-time	Full project	Ceremonies, risk, ZID dependency tracking
Peak Total	~9–10 FTE	—	—	Backend-only — frontend already complete

4.2 Three Team Size Options

Backend can ship with a smaller team over a longer timeline. Three viable configurations:

Option	Team Size	V1 Timeline	Composition	Tradeoff
Lean (Solo Founder)	5–6 FTE	7–8 months	Sole BE lead + 2 BE + 1 DevOps + 1 QA + fractional Sec/AI	Slower, sequential, no AI until later
Standard (Recommended)	9–10 FTE	5 months	Full pod with dedicated AI, Data, Sec	Balanced; on-time backend launch
Aggressive	13+ FTE	3.5–4 months	Double backend pod, dedicated infra team	Coordination tax, diminishing returns

Honest take:

9–10 FTE is right-sized for delivering full backend V1 in 5 months with parallel pods. If budget or hiring is constrained, the Lean 5–6 FTE option ships the same scope in 7–8 months by removing parallelism and deferring AI/Data work to later phases.

5. Module-wise ZID Integration Plan

Each module shows what ZID provides natively, where Haddiya adds custom logic, and the effort in person-weeks (PW).

#	Module	Description	ZID Coverage	Custom Effort	P W	Resources
1	ZID Partner App Registration	Register Haddiya as ZID partner app, scopes, OAuth credentials, sandbox + prod env config	Full (provided by ZID)	Config + flow wiring	2	1 BE Lead
2	OAuth 2.0 Store Connection	Per-vendor store authorization, token storage (Secrets Manager), refresh handling	Native OAuth flow	Token vault + refresh worker	4	1 BE, 1 Sec
3	ZID SDK / Connector Library	Typed NestJS client wrapping ZID REST API: retry, rate-limit aware (429 backoff), circuit breaker, request signing	REST endpoints only	All client logic in-house	6	1 BE
4	Webhook Ingestion Service	Receive ZID events (order.created, product.updated, customer.created, etc.), HMAC verify, idempotency, SQS dispatch	Webhook delivery only	Full ingestion + dedupe	8	1 BE
5	Catalog Sync Pipeline	ZID products → PostgreSQL + OpenSearch. Initial backfill + delta via webhooks. Bilingual fields, variants, media URLs	Products API (read)	Cache + reconciliation	10	1 BE
6	Inventory Sync	Real-time stock from ZID + reservation hold during checkout. 15-min reconciliation job	Stock webhook + API	Reservation layer + recon job	6	1 BE
7	Customer Sync	Mobile-first auth (phone+OTP) → ZID customer profile. Address book sync, GDPR/PDPL controls	Customer API	Auth bridge + privacy controls	8	1 BE, 1 Sec
8	Cart & Pricing Service	Multi-vendor cart, ZID pricing rules (tax, promotions), Haddiya-applied platform discounts	Cart API per store	Cross-store aggregation	10	1 BE
9	Order Orchestration	Haddiya checkout → 1..N ZID orders (one per vendor). Saga pattern for partial failures. Order status aggregation to customer	Order create API + webhooks	Saga + state machine	14	1.5 BE

#	Module	Description	ZID Coverage	Custom Effort	P W	Resources
10	Payment Integration via ZID	Leverage ZID gateway abstraction (Mada, STC Pay, Tabby, Tamara, Apple Pay, COD). 3DS handled by ZID	Full PSP coverage	Checkout glue + retry	6	1 BE, 1 Sec
11	Shipping & Carrier Integration	ZID carriers (SMSA, Aramex, Naqel) + Haddiya scheduled delivery windows stored as order metadata	Native carrier integrations	Schedule layer + tracking	8	1 BE
12	Refunds & Returns API	Initiate refund in ZID, sync vendor payout ledger, notify customer	Refund API	Payout reconciliation	6	1 BE
13	Multi-Vendor Mapping Service	Map Haddiya vendor → ZID store/sub-store. Vendor lifecycle (active, suspended, pending KYC)	ZID Marketplace plan or 1 store/vendor	Mapping + RBAC	12	1 BE
14	Vendor Payout Ledger	Track commission per order, split calc, payout schedule. Reconciles to ZID order totals daily	Order totals from ZID	Full ledger in Haddiya	12	1 BE, finance review
15	Vendor Onboarding API	KYC document upload (S3), approval workflow, ZID store provisioning via ZID admin API	Store provisioning API	KYC + approval flow	10	1 BE, ops
16	Vendor Portal API	Per-vendor scoped: products, orders, earnings, payouts. Built on ZID per-store APIs with Haddiya RBAC layer	ZID Products + Orders APIs	RBAC + earnings calc	14	1 BE
17	Gift AI Assistant Service	LLM + product embeddings (pgvector). Queries Haddiya catalog cache, never ZID directly	None (Haddiya-only)	Full build	12	1 AI, 1 BE
18	Send-as-Gift Backend	Message cards (templates), recipient details, gift wrapping options. Attached to ZID order via custom fields/metadata	Order metadata fields	Gift entity + assets	8	1 BE
19	Scheduled Delivery Logic	Date/time window selection, vendor capacity check, carrier handoff timing	Order metadata	Slot mgmt + validation	6	1 BE
20	Promotions & Coupons Bridge	Haddiya platform coupons + ZID native discounts. Stacking rules, eligibility validation	ZID discount API	Bridge + stacking logic	8	1 BE
21	Loyalty & Points Ledger	Earn on completed orders (ZID webhook), redeem via ZID coupon mint. Tiering rules	ZID coupon API	Full ledger + tiers	10	1 BE

#	Module	Description	ZID Coverage	Custom Effort	P W	Resources
2 2	Notifications Service	Push (FCM/APNs), WhatsApp Business, transactional email (SES). Subscribes to ZID order webhooks	Webhook events	All channels in-house	10	1 BE
2 3	Admin Operations API	Vendor approvals, dispute resolution, manual refunds, catalog moderation. Proxies to ZID where applicable	Admin endpoints	Workflow layer	10	1 BE
2 4	Reporting & Analytics Pipeline	ETL from ZID Orders API + Haddiya ledger → Snowflake. GMV, AOV, vendor performance, cohort retention	Orders/Customer APIs (export)	Full DW + dashboards	12	1 Data, 1 BE
2 5	Document Management Service	Vendor KYC, invoices (ZID provides), gift card art templates. S3 + signed URLs	Invoice generation	KYC + asset storage	4	1 BE
2 6	Customer Support API	Ticket flow, order lookup (ZID), refund actions, chat handoff (Zendesk/Intercom)	Order/refund APIs	Ticket layer	6	1 BE
2 7	Auth & RBAC	Haddiya users (customer/vendor/admin) — separate identity from ZID. JWT, refresh, MFA for vendors	ZID customer auth (consumer side)	Vendor/admin IdP + RBAC	8	1 BE, 1 Sec
2 8	Rate-Limit & Quota Mgmt	Stay within ZID API quotas (per-store + per-app). Distributed token bucket via Redis	ZID rate-limit headers	Quota orchestration	5	1 BE
2 9	Observability & Tracing	Track ZID API latency, webhook delivery success, retry counts. OpenTelemetry traces span Haddiya → ZID	ZID returns request IDs	Full instrumentation	8	1 DevOps
3 0	Security & Compliance	OWASP, PDPL, GDPR. ZID OAuth token vault, webhook HMAC, PII encryption at rest	ZID handles PCI scope	Haddiya-side hardening	10	1 Sec, DevOps
3 1	DevOps & Infrastructure	IaC (Terraform), CI/CD (GitHub Actions), ECS/EKS, env separation (dev/stage/prod), DR strategy	—	Full build	14	1 DevOps
3 2	Disaster Recovery & ZID Outage Playbook	Read-only mode when ZID down, queued writes, customer messaging, auto-replay when ZID recovers	—	Full design + drills	6	1 DevOps, 1 BE

Total: ~273 person-weeks across 32 modules.

6. Phase-wise Delivery Plan

Phase A — ZID Foundation · Weeks 1–4

- ZID partner app registration, OAuth credentials, sandbox environment.
- ZID API contract review, scope definition, rate-limit budget.
- Connector SDK skeleton (NestJS), authentication flow, retry/backoff.
- Webhook ingestion service, HMAC verification, SQS dispatch.
- PostgreSQL schema for cache, multi-vendor mapping, ledger tables.
- Initial CI/CD pipeline, staging environment with ZID sandbox.

Deliverables: Working OAuth flow, webhook ingest, ZID SDK v0.1, staging env.

Phase B — Core Commerce Sync · Weeks 4–10

- Catalog sync (products, categories, variants) → PostgreSQL + OpenSearch.
- Inventory sync via webhooks + 15-min reconciliation.
- Customer sync (Haddiya auth → ZID customer profile).
- Cart and pricing service (single-vendor first).
- Order orchestration v1 (single-vendor checkout, ZID order create, webhook listener).
- Payment integration via ZID gateway (Mada + COD initially).

Deliverables: End-to-end single-vendor purchase flow on staging. Webhook reliability > 99%.

Phase C — Multi-Vendor Layer · Weeks 8–14

- Vendor onboarding API + KYC workflow + ZID store provisioning.
- Multi-vendor mapping service.
- Cart splitting → N ZID orders, saga for partial failures.
- Vendor payout ledger + daily reconciliation against ZID totals.
- Vendor Portal API (products, orders, earnings scoped per vendor).

Deliverables: Two vendors transacting in parallel, accurate payout split.

Phase D — Differentiator Modules · Weeks 12–18

- Gift AI Assistant: embeddings pipeline, LLM-backed recommendation API.
- Send-as-gift backend (message cards, recipient details, wrapping).
- Scheduled delivery slot mgmt + carrier handoff timing.
- Loyalty ledger + ZID coupon bridge for redemption.
- Promotions bridge (Haddiya coupons + ZID discounts, stacking rules).

Deliverables: AI-driven recommendations live; gifting metadata flowing end-to-end.

Phase E — Operations & Ops APIs · Weeks 14–20

- Admin Operations API (approvals, disputes, refunds).
- Notifications service (push, WhatsApp, email) wired to ZID webhooks.
- Customer support API + ticketing integration.
- Refund and return flow with payout reconciliation.

Deliverables: Full admin workflows; ops team can manage vendors and orders without ZID UI.

Phase F — Analytics & Hardening · Weeks 16–22

- Reporting pipeline: ZID → Snowflake/BigQuery; Looker dashboards.
- Observability: OpenTelemetry traces, Datadog APM, RUM, error tracking.
- Security hardening: pen test, OWASP review, secrets rotation drill.
- Disaster recovery playbook + ZID outage simulation.
- Performance: k6 load tests at 5× expected peak.

Deliverables: Production-ready, monitored, security-cleared backend.

Phase G — Production Launch & Stabilization · Weeks 20–24

- Production cutover, blue-green deploy.
- On-call rotation, status page.
- Pilot vendor cohort (10–15) on production.
- Tuning based on real-traffic metrics.
- Backlog grooming for V2 (multi-country, loyalty deep-dive, B2B).

Deliverables: Live production with paying transactions. Runbooks complete.

7. Technical Architecture Stack

Layer	Recommendation	Rationale
Commerce Engine	ZID Platform (Marketplace plan)	GCC-native, PDPL-compliant, full payment + shipping coverage
Backend Framework	Node.js + NestJS (TypeScript)	Type-safe, modular, mature ecosystem
API Gateway / BFF	REST + tRPC for internal, GraphQL for mobile aggregation	Channel-tailored payloads
Database	PostgreSQL 16 on RDS (multi-AZ) + pgvector	ACID for ledger, embeddings for AI
Caching	Redis (ElastiCache)	Hot product cache, OAuth tokens, rate-limit buckets
Search	OpenSearch (managed) with Arabic + English analyzers	Catalog search, faceting, RTL tokenization
Queue / Events	Amazon SQS + EventBridge	Webhook fan-out, async order processing
Object Storage	S3 + CloudFront	KYC documents, gift card assets, signed URLs
Cloud Region	AWS me-central-1 (UAE) primary, eu-south-1 DR	PDPL data residency
Secrets	AWS Secrets Manager	ZID OAuth tokens, webhook HMAC keys, rotation
Observability	Datadog APM + logs + RUM, OpenTelemetry traces	ZID → Haddiya latency visibility
Error Tracking	Sentry	Full-stack exception capture
Feature Flags	LaunchDarkly or Unleash	Safe rollouts, vendor-segment targeting
Data Warehouse	Snowflake or BigQuery + Looker/Metabase	GMV, AOV, cohort, vendor performance
CI/CD	GitHub Actions → ECR → ECS Fargate	Blue-green, daily prod deploys
IaC	Terraform modules per env	Reproducible infra, drift detection
WAF / Security	AWS WAF + Shield	Bot mitigation, DDoS, OWASP top-10

8. Risk Analysis

8.1 Technical & ZID-Specific Risks

Risk	Likelihood	Impact	Mitigation
ZID API rate limits hit during catalog burst sync	Medium	High	Distributed token bucket, exponential backoff, off-peak backfill, request batching
Webhook delivery failures / out-of-order events	High	Medium	HMAC verify, idempotency keys, dead-letter queue, periodic reconciliation
ZID API contract changes (deprecation, breaking versions)	Medium	High	Pin API version, monitor ZID changelog, integration test suite, abstraction layer
Multi-vendor split semantics differ from ZID native model	High	High	Saga pattern, custom mapping table, daily reconciliation
OAuth token expiry during high-traffic windows	Low	High	Pre-emptive refresh worker, retry on 401, distributed lock on refresh

8.2 Scalability Concerns

Concern	Trigger	Mitigation
ZID API latency spikes under load	GMV spike (Eid, Mother's Day)	Aggressive caching, async order intake, queued writes
Webhook backlog during ZID incidents	ZID downtime > 5 min	SQS-backed buffer, replay on recovery, alerting
Catalog cache divergence from ZID	Catalog > 50K SKUs across vendors	Hourly reconciliation, drift metrics, alerting on > 0.5% variance
Payout ledger reconciliation lag	Daily order volume > 5K	Streaming reconciliation, materialized views, anomaly detection

8.3 Security Risks

Risk	Severity	Mitigation
ZID OAuth tokens leaked	Critical	Secrets Manager, no env vars, automated rotation, audit logging
Webhook spoofing (forged ZID events)	Critical	HMAC signature verification on every event, replay protection
Vendor data cross-tenant leak	High	Tenant scoping in every query, RLS on Postgres, isolation tests
PII exposure (Oman PDPL, GDPR)	Critical	Encryption at rest (KMS) + TLS 1.3, field-level PII, audit log, data residency in me-central-1
Supply-chain (npm) attack	Medium	Dependabot, SBOM, Snyk/Socket, signed images, lockfile audit

8.4 Performance Bottlenecks

Bottleneck	Symptom	Fix
ZID round-trips on every order page render	Slow product detail	Aggressive Redis cache, read-through from PG mirror

Bottleneck	Symptom	Fix
N+1 ZID calls during multi-vendor checkout	Slow checkout confirmation	Batch where API supports, parallel fan-out with bulkhead
Webhook processing slower than ingestion	Growing SQS backlog	Horizontal worker scale, batch consumption, prioritized queues
Reconciliation job blocks DB	Locks during peak	Read replica for recon, off-peak scheduling, chunked queries

9. Final Recommendation

9.1 Backend MVP Feature Set

Ship first:

- ZID partner app live with OAuth + webhook ingestion.
- Catalog sync (products + categories + variants) to Haddiya cache.
- Inventory sync (real-time webhooks + reconciliation).
- Customer auth bridge (phone+OTP → ZID customer).
- Single-vendor cart + checkout, ZID order create, ZID payment (Mada + COD).
- Order status webhooks → push notifications to customer.
- Basic admin API for order lookup + refund initiation.
- Vendor onboarding API (manual approval) + 1 store/vendor.
- Send-as-gift v1: message card text stored as order metadata.
- Observability baseline (logs + metrics + error tracking).

9.2 Defer to V1 / V2

Defer to V1:

- Multi-vendor cart splitting (saga + payout ledger).
- Gift AI Assistant (LLM + embeddings).
- Scheduled delivery slot management.
- Loyalty ledger + ZID coupon bridge.
- Vendor Portal API (self-service).
- Full payment mix (Tabby, Tamara, STC Pay, Apple Pay).
- WhatsApp Business notifications.
- Reporting pipeline + warehouse.

Defer to V2:

- Multi-country expansion (UAE, Bahrain ZID stores) with multi-currency.
- B2B corporate gifting API (bulk orders, invoicing, custom branding).
- Subscription gifting / recurring deliveries.
- Advanced anomaly detection (vendor fraud, inventory anomalies).
- Multi-warehouse routing logic.
- Marketplace ads / sponsored placement API.
- Vendor mobile app backend.

9.3 Best Timeline & Milestones

Milestone	Calendar Week	Outcome
ZID partner app + OAuth working	W 2	Sandbox transactions possible
Catalog + inventory sync live	W 6	Cache mirrors ZID in real time
Single-vendor checkout end-to-end	W 10	MVP backend launches private beta
Multi-vendor split + payout ledger	W 14	Marketplace flow functional
Gift AI + send-as-gift backend	W 18	Differentiators live
V1 production launch	W 22	Public consumer launch alongside frontend
V2 GCC expansion start	W 26	Second ZID region onboarded

9.4 Fastest Realistic Approach

- **Pin ZID API version.** Read changelog weekly. Integration test suite runs nightly against sandbox.
- **Webhook-first, never poll.** Polling at scale exhausts rate limits. Reconciliation as safety net only.
- **Idempotency keys everywhere.** Every ZID write carries deterministic external_id derived from Haddiya domain event.
- **Cache aggressively, invalidate on webhook.** Hot products + categories cached in Redis with 15-min TTL + webhook bust.
- **Saga + dead-letter queues for multi-vendor orders.** Partial failures must be observable and replayable.
- **Daily payout reconciliation in dev from day one.** Build the financial truth-table before any vendor ships.
- **Sandbox-driven development.** Every engineer has a personal ZID sandbox; no shared dev env coupling.
- **Contract tests against ZID API.** Pact or schema snapshots catch ZID changes before they reach prod.
- **Pre-book ZID partner support contact.** Direct escalation line for incidents and API questions.
- **Reserve 15% capacity for ZID quirks.** GCC commerce platforms have edge cases — VAT, address formats, name fields.

Bottom line:

A focused backend team of 9–10 FTE delivers production-ready ZID-integrated backend V1 in ~5 months. ZID provides the commerce engine; Haddiya engineers build the marketplace, AI, and gifting differentiators on top. Backend launches alongside the already-completed frontend.

Document prepared for executive review. API specifications, sprint backlog, and ZID partner contract details available as appendices.